Christian Marowski

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Education

Indiana University | Bloomington, Indiana *Aug 2025 – May 2027*

Master of Science in Computer Science

- GPA – 3.78

Relevant Courses: Multi-Robot Systems, Robot Kinematics, Robotics Algorithms, Signal Processing, Computer Vision

Marquette University | Milwaukee, Wisconsin *December 2024*

Bachelor of Science

- Majors in Computer Science and Cognitive Science
- GPA – 3.49
- Member of UPE Computer Science Honor Society

Relevant Courses: Hardware Systems, Programming Languages, Cognition, Neuroscience, Fundamentals of Artificial Intelligence, Epistemology

Publications

1. **Christian Marowski** and Byung-Cheol Min. **"Task Structure and Scalability in Multi-Robot Authoring,"** *Submitted to the 2027 IEEE International Conference on Robotics & Automation (ICRA)* (under review).
2. S. G. Ramagoni, **C. Marowski**, A. Christie, B. Cox, D. Brylow, and A. G. Bourgeois. **"A Comparative State-Level Analysis of Computer Science Teacher Capacity in Georgia and Wisconsin Public High Schools,"** *2025 IEEE Frontiers in Education Conference (FIE)*, Nashville, TN, USA, 2025, pp. 1–9.

Research Experience

Research Assistant *May 2026 – Present*

SMART Lab (Smart Machine and Robotics Technology)
Indiana University | Dr. Byung-Cheol Min

- Investigate multi-robot coordination using classical and learning-based approaches to cooperative autonomy and decision-making.
- Developed a multi-robot authoring simulator and physical-robot evaluation tooling; conducted 4,620 simulation runs across 385 drawings and 6 team sizes.
- Explore graph-based reinforcement learning and multi-agent credit assignment to improve coordination and adaptability in multi-robot systems.

Optimization Team Member *November 2025 – May 2026*

LART (Luddy Autonomous Racing Team)
Indiana University | Dr. Lantao Liu, Hassan Jardali

- Contributed to camera and LiDAR sensor redundancy strategies to improve the reliability of the vehicle's perception system.
- Refined Git branching and version control workflows to support integration across the autonomous racing team.

Data Research Contractor *August 2025 – January 2026*

Wisconsin Department of Public Instruction (DPI)
Milwaukee, Wisconsin

- Transformed raw Wisconsin public school data into a standardized format according to ECEP Alliance Common Metrics Project requirements using Python scripts.
- Authored technical documentation detailing data cleaning, mapping, and transformation logic.
- Coordinated with Wisconsin DPI, the Common Metrics Project team, and Marquette research group to align project goals and deliverables.

Research Experience (continued)

Research Assistant *Summer 2024 – August 2025*

PISCES (Promoting Innovative Systems for Computing Education in Schools)
Marquette University | Dr. Dennis Brylow

- Processed and standardized 7 years of statewide Wisconsin public high school data, encompassing approximately 250,000 students and 2 million course records, to support large-scale educational research.
- Developed Python-based data normalization and analysis methods to prepare datasets.
- Designed interactive dashboards using Plotly and Tableau to visualize statewide educational trends and communicate research findings.

Teaching Experience

Head TA – Operating Systems *Jan 2025 – May 2025*

Marquette University | Dr. Dennis Brylow

- Led instructional support for an operating systems course emphasizing hands-on implementation of core components, including scheduling, memory management, and synchronization.
- Coordinated teaching assistants, conducted office hours, and guided students through programming assignments, debugging, and systems-level concepts.

Projects

Reinforcement Learning for Multi-Robot Formation Control *June 2026 – Present*

Ongoing Research | SMART Lab / Master's Thesis

- Developing an end-to-end simulation-to-real framework integrating Webots robot models, physical robot platforms, and improved vision-based localization.
- Investigating graph-based multi-agent reinforcement learning and credit assignment to improve formation maintenance and recovery from disruptions.
- Designing simulated and physical experiments to evaluate learned and classical controllers on formation accuracy, collision avoidance, and recovery performance.

Vision-Based Multi-Robot Coordination Tutorials *May 2026 – Present*

Research Project | SMART Lab

- Developed a reusable multi-robot control framework integrating camera-based localization, ArUco marker tracking, and differential-drive robot actuation.
- Implemented formation control and flocking algorithms for physical robot teams using real-time visual pose feedback.
- Developing reproducible tutorials documenting controller implementations, mathematical derivations, and coordination strategies.

GNN for Flocking Control *March 2026 – May 2026*

Multi-Robot Systems | Class Project

- Designed and implemented a Graph Neural Network (GNN) to learn multi-robot flocking behaviors through imitation of classical controllers.
- Developed a Webots simulation framework to generate graph-structured training data and evaluate learned control policies for a team of 6 robots.
- Achieved comparable average formation error (0.013 vs. 0.012) and cycle completion time (493 vs. 488 seconds) to classical control, while observing 3 collisions compared to 0 for the baseline.

Skills

- **Programming Languages:** Python, C/C++, Java, JavaScript, SQL, R, Assembly
- **Robotics & Perception:** Computer Vision, Robot Localization, Pose Estimation, Sensor Processing, Sensor Integration
- **Planning & Control:** Multi-Robot Coordination, Motion Planning, Task Allocation, Robot Kinematics, Feedback Control, Numerical Optimization
- **Machine Learning:** Deep Learning, Reinforcement Learning, Graph Neural Networks, Imitation Learning, Supervised Learning
- **Tools & Frameworks:** PyTorch, Webots, OpenCV, SciPy, NumPy, Pandas, Matplotlib, Git, LaTeX